



Q1. What are the common types of motors and batteries used in two-wheeler electric vehicles (EVs) in India?

Abinaya V., Coimbatore

A1. Like any other vehicle, EVs have different specifications and features depending on the models and variants. Actual specifications of a particular model can be obtained from the manufacturer's or authorised dealer's website. For example, Avon e-bike, a two-wheeler electric bicycle, is powered by a 137W brushless DC electric motor and three 12V (3x12V = 36V), 12Ah rechargeable lead acid batteries. The batteries can be charged via 220V AC, which takes four to eight hours to fully charge them. With a fully-charged battery, this e-bike can run at a maximum speed of 20-23kmph. Details of some EVs taken from the dealer's website are listed in the table.

Different Types of Electric Bikes

Type of electric vehicle	Motor type	Power	Battery type	Charging time (in hours)	Range (km/charge)
Bicycle (Avon e-bike)	Brushless DC	137W	36V, 12Ah rechargeable battery	4-8	40
Scooter (Bajaj Chetak Urbane)	Brushless DC	4080W	Lithium-ion	5	95
Bike (Revolt RV300)	Brushless DC	1.5KW	Lithium-ion	4.2	180

Bajaj Chetak e-scooter from Bajaj Auto uses an electric motor rated at 4kW of peak power, 3.8kW of continuous power and 16Nm of torque. It has two operational modes, namely, Eco and Sport. In Eco mode, Chetak has a claimed maximum range of over 95km, while in Sport mode, its range is limited to over 85km on a single charge. According to the dealer's website, battery life is about 70,000km and it can be charged up to hundred per cent in five hours.

RV300 e-bike from Revolt Motors is currently available for sale in Pune and Delhi only. It uses lithium-ion battery and 1.5kW hub DC motor. The bike comes with a claimed range of 80km to 180km on a full charge.



Fig. 1: Typical lithium-ion battery pack for e-bike

Q2. What is the procedure for programming ESP32 microcontroller (MCU) using different platforms? Which operating system (OS) is used for ESP32? What are the features and applications of ESP32 MCU?

Pamarthi Kanakaraja, Andhra Pradesh

A2. There are many options to program ESP32, including Arduino programming language using Arduino IDE, Python programming using MicroPython, Lua-based firmware using NodeMCU IoT (Internet of Things) platform, JavaScript or C programming using Mongoose OS.

Mongoose OS is one of the programming platforms for ESP32 MCU recommended by Espressif Systems. ESP32 is the successor to ESP8266 and, hence, has Wi-Fi and Bluetooth support. It is perfect for any IoT project. It has 802.11b/g/n HT40 Wi-Fi transceiver, so it can connect to a Wi-Fi network, and can also set up a network of

its own, allowing other devices to connect directly to it.

It also supports Wi-Fi Direct for peer-to-peer connection without the need of an access point. The chip also has dual-mode Bluetooth capabilities with both Bluetooth 4.0 (BLE/Bluetooth Smart) and Bluetooth Classic (BT), suitable for versatile wireless applications.

For a sample application, you may refer to 'The IoT And Touch-Based Home Automation using ESP32' DIY project from our website, www.electronicsforu.com